

Diagnostic information - amplifier

Front left/right loudspeaker - tweeter - short circuit between lines

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance **< 0.5 ohm**

Possible fault cause(s)

- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Subwoofer - acoustic error

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Subwoofer self-test not OK

Possible fault cause(s)

- Line faulty
- Lines and/or connectors corroded
- Internal fault in subwoofer
- Implausible signal

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

000291

Diagnostic information - amplifier

Subwoofer - polarity reversed

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Subwoofer self-test not OK

Possible fault cause(s)

- Polarity reversed
- Line faulty
- Lines and/or connectors corroded
- Internal fault in subwoofer
- Implausible signal

Fault effect(s)

- Subwoofer has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Rear left/right loudspeaker - tweeter/mid-range - acoustic error

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Loudspeaker self-test not OK

Possible fault cause(s)

- Line faulty
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker has limited functionality or does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Rear loudspeaker - polarity reversed

Diagnostic conditions

- Ignition on
- PCM on
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- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Loudspeaker self-test not OK

Possible fault cause(s)

- Polarity reversed
- Line faulty
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker has limited functionality or does not work
- Changed sound

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Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

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Fault effect(s)

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Setpoint values for line test

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Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

000600

Diagnostic information - amplifier

Amplifier supply voltage above limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too high

Possible fault cause(s)

- Supply voltage too high

Fault effect(s)

- Amplifier has limited functionality

000601

Diagnostic information - amplifier

Amplifier supply voltage below limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too low

Possible fault cause(s)

- Supply voltage too low

Fault effect(s)

- Amplifier has limited functionality

000602

Diagnostic information - amplifier

Amplifier supply voltage above limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too high

Possible fault cause(s)

- Supply voltage too high

Fault effect(s)

- Amplifier has limited functionality

000603

Diagnostic information - amplifier

Amplifier supply voltage below limit value

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier detects a supply voltage that is too low

Possible fault cause(s)

- Supply voltage too low

Fault effect(s)

- Amplifier has limited functionality

000606

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000607

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000608

Diagnostic information - amplifier

Amplifier - programming

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect programming

Possible fault cause(s)

- Programming incorrect or not carried out
- Software incompatible
- Amplifier faulty

Fault effect(s)

- Various function restrictions

000614

Diagnostic information - amplifier

Amplifier - programming

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect programming

Possible fault cause(s)

- Programming incorrect or not carried out
- Software incompatible
- Amplifier faulty

Fault effect(s)

- Various function restrictions

000617

Diagnostic information - amplifier

Amplifier - fan, faulty

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Fan on > 2 minutes

Fault setting condition(s)

- Amplifier detects an electrical fault in the internal fan

Possible fault cause(s)

- Overtemperature protection active
- Open circuit in wiring
- Short circuit to ground/B+
- Fan faulty

Fault effect(s)

- Housing fan does not work or has limited function

000631

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000633

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000635

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000636

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000640

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000660

Diagnostic information - amplifier

Active noise reduction - engine speed - implausible signal

Diagnostic conditions

- Ignition on
- Engine speed > **1,000 rpm**
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier receives an implausible engine speed signal of < **500 rpm**

Possible fault cause(s)

- Signal implausible
- Line faulty
- Lines and/or connectors corroded
- Amplifier faulty

Fault effect(s)

- Active noise reduction limited or does not work

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V** ⇒ **+ 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance □ **0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance < **2 ohm**

000661

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000662

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000663

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

000664

Diagnostic information - amplifier

Amplifier - coding

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Amplifier reports incorrect coding

Possible fault cause(s)

- Coding incorrect or not carried out

Fault effect(s)

- Various function restrictions

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

001007

Diagnostic information - amplifier

Authentication failed - signature/certificate

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on

Fault setting condition(s)

- Data transmission of the MOST components is encrypted using a protocol. The encryption cannot be verified by the communication partner.

Possible fault cause(s)

- The signature of the MOST control unit cannot be verified
- The communication certificate is not correct

Fault effect(s)

- No or restricted function of the relevant MOST component

Diagnostic information - amplifier

Subwoofer - open circuit

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance > **20 ohm**

Possible fault cause(s)

- Open circuit
- Lines and/or connectors corroded
- Internal fault in subwoofer
- Implausible signal

Fault effect(s)

- Subwoofer does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range/woofer - open circuit

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance > **20 ohm**

Possible fault cause(s)

- Open circuit
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range/woofer - short circuit to ground/B+

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance **< 3 ohm**

Possible fault cause(s)

- Short circuit to ground/B+
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range/woofer - short circuit to ground/B+

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance **< 3 ohm**

Possible fault cause(s)

- Short circuit to ground/B+
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Front left/right loudspeaker - mid-range/woofer - short circuit between lines

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance **< 0.5 ohm**

Possible fault cause(s)

- Short circuit between lines
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
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Open circuit:

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Diagnostic information - amplifier

Front left/right loudspeaker - mid-range/woofer - open circuit

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance > **20 ohm**

Possible fault cause(s)

- Open circuit
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
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Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Rear left/right loudspeaker - tweeter - open circuit

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance > **35 ohm**

Possible fault cause(s)

- Open circuit
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

Setpoint values for line test

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V ⇒ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **□ 0.9 Mohm**

Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Front left/right loudspeaker - tweeter - short circuit to ground/B+

Diagnostic conditions

- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance **< 3 ohm**

Possible fault cause(s)

- Short circuit to ground/B+
- Lines and/or connectors corroded
- Loudspeaker diaphragm faulty
- Internal fault in loudspeaker
- Implausible signal

Fault effect(s)

- Loudspeaker does not work
- Changed sound

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Open circuit:

- ♦ Setpoint resistance **< 2 ohm**

Diagnostic information - amplifier

Front left/right loudspeaker - tweeter - short circuit to ground/B+

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- Ignition on
- PCM on
- Amplifier on
- Battery positive voltage **10.8 V - 16 V**

Fault setting condition(s)

- Load resistance **< 3 ohm**

Possible fault cause(s)

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Fault setting condition(s)

- Load resistance **< 0.5 ohm**

Possible fault cause(s)

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- Ignition on
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- Load resistance > **35 ohm**

Possible fault cause(s)

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